

Leveraging Large Language Models and Machine Learning to Identify Inconsistencies in Electronic Health Records

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ABSTRACT

Objective: Discrepancies in the intended dosage of a drug and the actual administered dose can lead to severe consequences. Extracting intended dosages from clinical notes and validating them against the recorded doses in structured data can help alert caregivers of such errors.

Materials and Methods: This project uses the Medical Information Mart for Intensive Care III (MIMIC-III) dataset. Text is cleaned with the Natural Language Toolkit (NLTK) and Gemini 2.0 Flash serves as the large language model (LLM) for dosage extraction. Data is structured using Pydantic, embeddings are generated with text-embedding-004, and various classifiers—consisting of AdaBoost, Gradient Boosting, Decision Tree, Logistic Regression, K-Nearest Neighbors, Neural Network, Naive Bayes, Quadratic Discriminant Analysis, and Random Forest—are used for prediction tasks.

Results: The initial logistic regression provided AUROC of 57.65 and AUPRC of 8.16%. For the initial Random Forest model, the F1-score was 100% for false results and 56% for true. The rest of the classifiers performed extremely well across most metrics with accuracy and F1-score above 98%. Gradient Boosting, AdaBoost, Random Forest, K-Nearest Neighbors, and Naive Bayes were excellent for ROC AUC, but the other classifiers scored lower in this metric.

Conclusions: The findings from this study highlight the potential LLMs and machine learning (ML) have to solve issues like medication errors and lay the groundwork to build upon the solution.

KEYWORDS

Large Language Models, Machine Learning, Electronic Health Records, Medication Errors, Dosage Discrepancies, Healthcare AI

1 INTRODUCTION

Medication errors remain a prevalent—and sometimes fatal—issue in the healthcare industry. In the tragic case of Betsy Lehman, a misinterpretation of a chemotherapy dosage order led to her receiving four times the intended amount, ultimately resulting in her death [1]. Such incidents highlight the crucial need for safeguards against errors in medical documentation. Clinical notes often lack standardization since each medical professional has a unique style of thinking and documenting medical information. As

a result, it is challenging for other medical professionals to consistently extract and validate prescribed dosages, which is crucial when administering the medicine. Given the advancements in natural language processing (NLP) and LLMs, information extraction from free-text clinical notes is becoming a more promising solution. The majority of existing research focuses on identifying adverse drug events (ADEs). Validation of correct drug dosage, however, has received relatively less attention despite the tragic cases such as Betsy Lehman’s incident. This is a crucial gap to research as administering incorrect doses could potentially be just as dangerous as ADEs. This project aims to automatically detect discrepancies between prescribed dosages and administered dosages by comparing free-text clinical notes with structured medication administration records.

2 RELATED WORK

Recent advancements in NLP and LLMs have prompted more studies on the use of these technologies in clinical information extraction. Prior studies have explored both traditional and modern approaches to extracting data from free-text clinical notes, with a particular focus on general medication information. This section highlights key contributions to this healthcare challenge, as well as research gaps that motivate the direction of this study.

2.1 Information Extraction from Clinical Notes: Are We Ready to Switch to Large Language Models?

Hu et al. [2] examined whether LLMs are ready for information extraction (IE) from clinical notes. Their approach involved analyzing the performance of instructing LLaMA-2 and LLaMA-3 to extract problems, tests, medications, and treatments from clinical notes. The corpus was built on data from UT Physicians (UTP), MTSamples, MIMIC-III, and i2b2. Ultimately, the study found that while the LLaMA models consistently outperformed the BiomedBERT baseline, they also consistently introduced significantly higher computational demands compared to BERT-based models. A key contribution from this study was the development of Kiwi, their publicly available LLM-based clinical IE system that allows users to select tools based on their task requirements and computational resource constraints. While this work showcases the potential of LLMs for

clinical contexts, it does not delve into validation of the extracted information against structured records.

2.2 Medication Information Extraction using Local Large Language Models

Another study by Richter-Pechanski et al. [3] evaluated local LLMs for extraction of medical information. LLaMA and OpenBioLLM were the two LLMs selected to be evaluated on English and German corpuses. Similar to the previous study, the fine-tuned LLaMA models performed well. While this work advances the application of LLMs for medication information extraction, it again does not address the validation of extracted medication dosages against structured administration records, which is the gap that this study aims to address.

2.3 Building Longitudinal Medication Dose Data Using Medication Information Extracted from Clinical Notes in Electronic Health Records

Finally, McNeer et al. [4] aimed to address the challenge of "building longitudinal medication dose datasets [by extracting medication-related information] from clinical notes in electronic health records (EHRs)." To achieve this, the biostatistics and biomedical experts developed an NLP algorithm to process clinical notes to extract medication doses. They then utilized data from Vanderbilt University's EHR system and MIMIC-III to validate the performance of the algorithm. The algorithm performed relatively well with an F1-measure greater than 0.98 for both dose intake and daily dose in internal validation. The external validation had F1-measure of 0.85 for dose intake and 0.82 for daily dose. This work was an impressive development for improving automated extraction from medical notes, specifically for drug dosage. Its work both provides this study with valuable insights on using NLP to extract dose data from clinical notes and leaves a gap to explore cross-validation between those extractions and administration records.

3 METHODOLOGY

This section describes the dataset selected for the project, the process of choosing the medication and tables for analysis, and the final systematic approach that was followed (Figure 1).

3.1 Dataset

The dataset chosen for this study was MIMIC-III developed by the MIT Laboratory for Computational Physiology [5]. MIMIC-III contains a wide range of de-identified health records. This project specifically utilizes the patients, admissions, d_items, inputevents_mv, and noteevents tables. The inputevents_mv table contains structured information on administered drug doses recorded by the MetaVision system, while the noteevents table contains free-text clinical notes. These two tables served as the primary sources for comparing recorded drug administrations against descriptions in clinical notes.

3.2 Drug and Input Selection

In order to ensure a meaningful analysis, it was critical to identify the specific medication(s) and corresponding input events to study. This section details the process that went into both selections.

3.2.1 Selecting the Medication. Given the wide variety of drugs recorded in MIMIC-III, there was a higher priority on the medications that are frequently prescribed and possess high risk if an incorrect dose is administered. An initial list of Insulin, Fentanyl, Vicodin, Adderall, Morphine, Xanax, and Percocet—among others—were generated with this criteria in mind. Due to the amount of resources required to investigate the full list of drugs, however, it was decided to focus on a single drug for the scope of this project. The first choice was Insulin which has abundant data and is known to be dangerous when improperly dosed. However, during data cleaning, there were many inconsistencies found in the way Insulin doses were recorded—both in the clinical notes and the structured data. Particularly, many of the rows were unclear on the unit of measurement for the dose, such as "UNIT". Because of these challenges, the team shifted focus to Fentanyl. Fentanyl also has plenty of entries in MIMIC-III but unlike Insulin, exhibits much more consistency in the unit of measurement. Almost all of the entries are recorded in some fraction of grams (micrograms [mcg], milligrams [mg], etc). Given the availability of data and the consistency in recording, Fentanyl was selected as the drug of focus for this study.

3.2.2 Selecting the Input Events. Another important decision was selecting which table to use for the recorded dose administered to the patient. The two tables in MIMIC-III that include this data are inputevents_cv (CareVue) and inputevents_mv (MetaVision). Ultimately, the MetaVision table was selected for several reasons. First, MetaVision is the more modern system compared to CareVue. It also provides more structured and standardized data while CareVue exhibits more data inconsistencies. Lastly, after merging input event data with the rest of the dataset, inputevents_mv offered more usable records—likely due to the improved structure and adoption of MetaVision in later hospital records.

3.3 Final Workflow

3.3.1 Data Cleaning and Preprocessing. To begin the analysis, Google Drive was mounted, and the MIMIC-III data, originally downloaded from PhysioNet, was loaded into Pandas DataFrames from various CSV files. Input events related to the drug "Fentanyl" were extracted from MetaVision. Clinical notes from the NoteEvents table were filtered to retain only relevant categories and mentions of the drug. For further processing, NLTK resources were employed to clean and preprocess the clinical notes. This involved a series of text processing steps, including converting text to lowercase, removing unwanted content such as punctuation and URLs, tokenizing, and optionally applying stopword removal, stemming, or lemmatization. Additionally, a proximity search was conducted to identify instances of drug names and dosages within specified character ranges (e.g., 25 characters) from each other. We proceeded by merging clinical notes and input events using key identifiers such as SUBJECT_ID (patient), HADM_ID (admissions), and CGID (caregiver). To ensure the integrity of the merged data, filtering was applied to confirm that chart times in the NoteEvents table

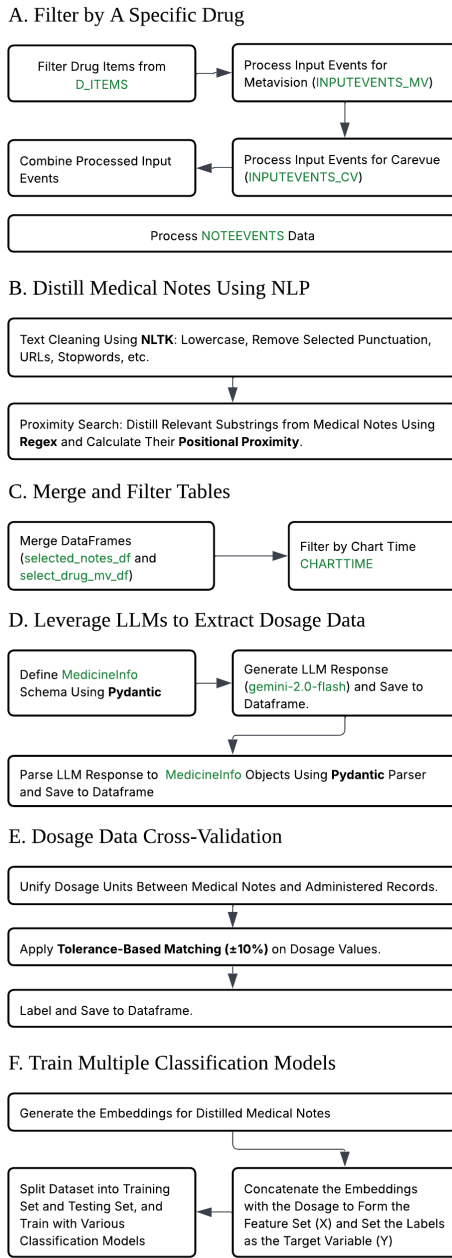


Figure 1: Flowchart representing workflow of the project

fell within the start and end times of the input events. Extensive manual testing was conducted to verify the accuracy of the data and confirm that the resulting dataset met the requirements for further analysis.

3.3.2 Google Gemini Gen AI API, Pydantic and LangChain. We leveraged the Google Gemini AI API to obtain structured responses from the "gemini-2.0-flash" model. Gemini 2.0 Flash is a next-generation multimodal model that can process text, images, audio, video, and

code. It offers superior speed, built-in tool use, and a 1-million-token context window. It features Grounding with Google Search for improved accuracy, code execution and function calling, batch prediction and controlled generation, and multimodal generation (text, images, audio). It is designed for low latency and enhanced performance, making it suitable for real-time applications. The Pydantic Output Parser is a class from the LangChain library that allows you to parse outputs from large language models (LLMs) using a predefined schema based on Pydantic models. Pydantic is a Python library for data validation and settings management using Python type annotations. It allows you to specify the structure and types of data, and automatically validates the input against the defined schema, hence using the Pydantic BaseModel, the LLM responses were formatted to include structured details about medicines and their dosage units. The DataFrame was updated with these responses, followed by validating dosage matches within a 10% threshold for accuracy. Given the time-intensive nature of querying the LLM, batch processing was implemented to streamline data handling and prevent overwhelming the LLM with excessive requests in a short timeframe.

3.3.3 Gen AI Embeddings, Logistic Regression model and Random Forest Classifier. A specialized function was developed to generate embeddings for semantic representation using the Gemini model "models/text-embedding-004," which converts text into dense embeddings optimized for semantic understanding, resulting in outputs of dense vectors that can be used with similarity measures like cosine similarity or dot product. The task_type config used was "RETRIEVAL_QUERY," which is optimized for search queries with the task type. These embeddings are designed to match queries with relevant documents in a retrieval system. A random subset of data was extracted from the main DataFrame and partitioned into training and testing sets in a 67:33 ratio. The training data was first used to build a Logistic Regression model, followed by the implementation of a Random Forest Classifier model. To evaluate the performance of these models, ROC AUC scores were computed for each class, providing valuable insights into the metrics and overall effectiveness of the classifiers.

3.3.4 Multiple Classification Models Final Evaluation. A range of classification-based machine learning models were utilized, including AdaBoost with the SAMME algorithm, Gradient Boosting configured with 300 estimators, Decision Tree with a maximum depth of 5, Logistic Regression allowing up to 1000 iterations, K-Nearest Neighbors with 5 neighbors, Neural Network with an alpha value of 1, Gaussian Naive Bayes, Quadratic Discriminant Analysis (QDA), and Random Forest Classifier using 300 estimators. These models were trained on embeddings generated from the dataset, with a 67:33 train-test split. Performance metrics such as precision, recall, and ROC AUC scores were calculated for each classifier. The results were documented in a statistics DataFrame, enabling visualization and comparison of the models' performance.

4 RESULTS

4.1 Initial Logistic Regression Metrics

A Logistic Regression model is a statistical method used for binary classification tasks, where the goal is to predict one of two possible

outcomes (e.g., "yes" or "no," "success" or "failure"). Unlike linear regression, which predicts continuous values, logistic regression predicts probabilities and maps them to discrete classes using a mathematical function called the sigmoid function. The results were AUROC of 57.65 and AUPRC of just 8.16%.

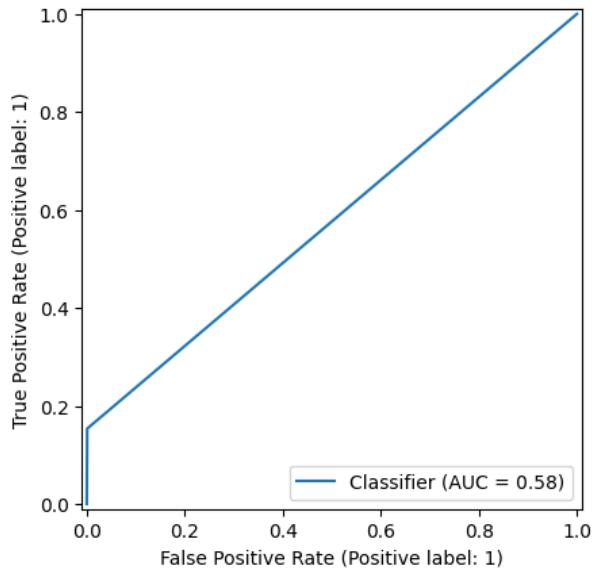


Figure 2: Receiver Operating Characteristic (ROC) curve of initial Logistic Regression

Accuracy Score: 0.9945008460236887				
Classification Report:				
	precision	recall	f1-score	support
False	1.00	1.00	1.00	2351
True	0.50	0.15	0.24	13
accuracy			0.99	2364
macro avg	0.75	0.58	0.62	2364
weighted avg	0.99	0.99	0.99	2364
ROC AUC Score: 0.5764977260085724				

Figure 3: Logistic Regression performance metrics

4.2 Initial Random Forest Metrics

These are the most popular machine learning algorithms that belong to the family of ensemble methods. It works by combining multiple decision trees to make predictions and improve accuracy. The algorithm is particularly effective for both classification and regression tasks and excels at handling large datasets with complex structures. The F1-score with random forest classifier model was reported as 100% for false results for dosage matching between input events and notes events. True results, however, had an F1-score of 56%.

	precision	recall	f1-score	support
False	1.00	1.00	1.00	2351
True	1.00	0.38	0.56	13
accuracy			1.00	2364
macro avg	1.00	0.69	0.78	2364
weighted avg	1.00	1.00	1.00	2364
Average ROC AUC Score: 0.9611376500997939				

Figure 4: Random forest performance metrics

4.3 All Classifiers Metrics

All of the classifiers achieved great scores for precision, recall, accuracy, and F1 with at least 98% for all those categories. Gradient Boosting, AdaBoost, Random Forest, K-Nearest Neighbors, and Naive Bayes all surpassed the 98% mark for ROC AUC. Decision Tree, Logistic Regression, and Neural Network were slightly worse with 70.74%, 94.02%, and 85.9% respectively. Quadratic Discriminant Analysis (QDA) performed the worst with 54.88%.

	Precision	Recall	Accuracy	F1	ROC AUC
Gradient Boosting	1.000000	1.000000	1.000000	1.000000	1.000000
AdaBoost	0.999792	0.999792	0.999792	0.999790	0.999992
Random Forest	0.997297	0.997290	0.997290	0.997037	0.999990
K-Nearest Neighbors	0.994489	0.994997	0.994997	0.994571	0.997559
Decision Tree	0.995022	0.994997	0.994997	0.993968	0.707379
Logistic Regression	0.992552	0.992495	0.992495	0.989573	0.940158
Neural Network	0.992346	0.992287	0.992287	0.989131	0.859038
QDA	0.992346	0.992287	0.992287	0.989131	0.548780
Naive Bayes	0.993172	0.986658	0.986658	0.989150	0.989320

Figure 5: A classification models' performance metrics

5 CONCLUSION

The purpose of this study was to utilize large language models and machine learning to find discrepancies between medication dosages from free-text clinical notes and recorded doses in structured data. The results from the work were promising and laid groundwork for future improvements upon the findings.

5.1 Key Findings

The accuracy that the classification models had were extremely promising for "false" results. Many of the models achieved accuracies and F1-scores greater than 99%. However, the samples—and dataset in general—had very few "true" labels, which leaves room for improvement on such data.

5.2 Real-World Application

The findings from this study can be extremely valuable for identifying discrepancies between doses in free-text clinical notes and structured administered dose records. Once identified, caregivers can be alerted of dosage errors and act accordingly. These inconsistencies can lead to serious consequences such as those in the Besty Lehman case. The goal is to diminish the frequency of such cases by utilizing technologies like large language models and machine learning.

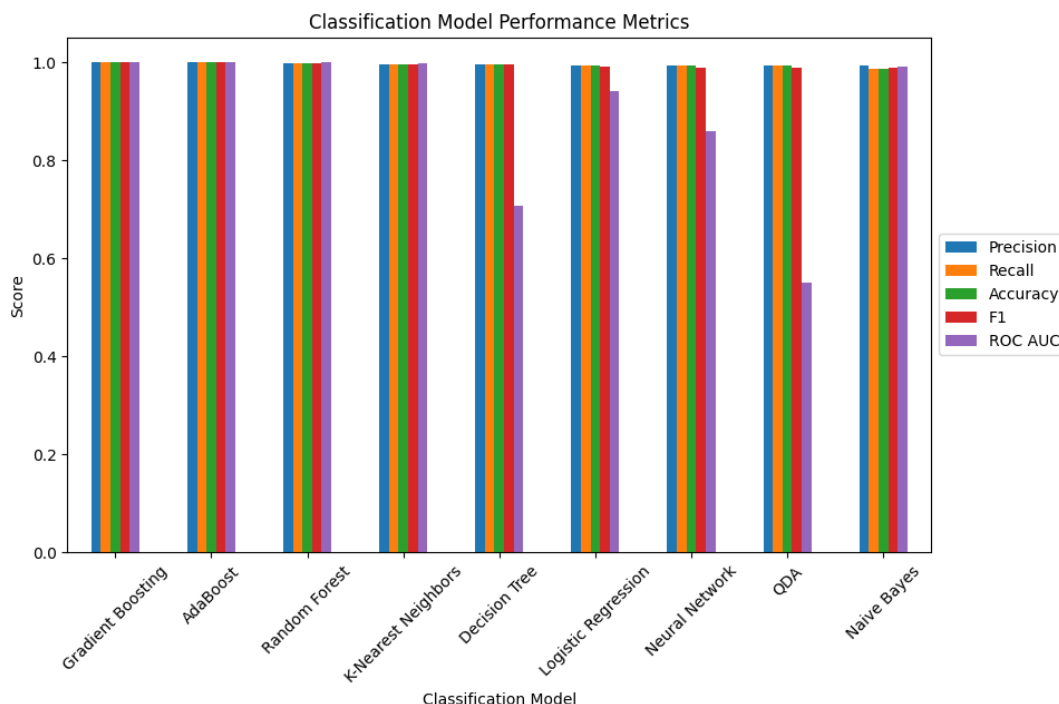


Figure 6: Bar graph representing classification models' performance metrics

5.3 Future Work

5.3.1 Expanding the Dataset. The original plan for this project included exploring multiple high-risk medications where dosage errors could lead to serious consequences. Expanding the dataset to include more such medications would further enrich the extraction and training process, improving model robustness and generalizability. Given the low quantity of matches between the prescriptions and the administered doses, the data was very skewed. Generating synthetic matches could be a potential avenue for remedying this ratio.

5.3.2 More LLM Reasoning Methods. This study primarily utilized zero-shot prompting for the LLM. Introducing techniques such as few-shot learning, chain-of-thought prompting, and tree-of-thought reasoning could significantly enhance the quality and validity of the LLM extractions.

5.3.3 Model Refinement. While the current model performs well in identifying discrepancies between extracted and structured dosage information, precision and accuracy are lower when the extracted dosage matches the structured data. This is likely due to the limited amount of "true" matches that exist in the dataset. Dataset expansion, extensive prompting strategies, and potential fine-tuning of the models could address this limitation.

5.3.4 Testing Model on Unseen Data. Thus far, the models have only been evaluated on the thirty percent of the sample designated as test data. Extending evaluation to truly unseen data—such as

datasets outside of MIMIC-III—would provide a better understanding on how well the models work and reveal additional avenues for improvement. a

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